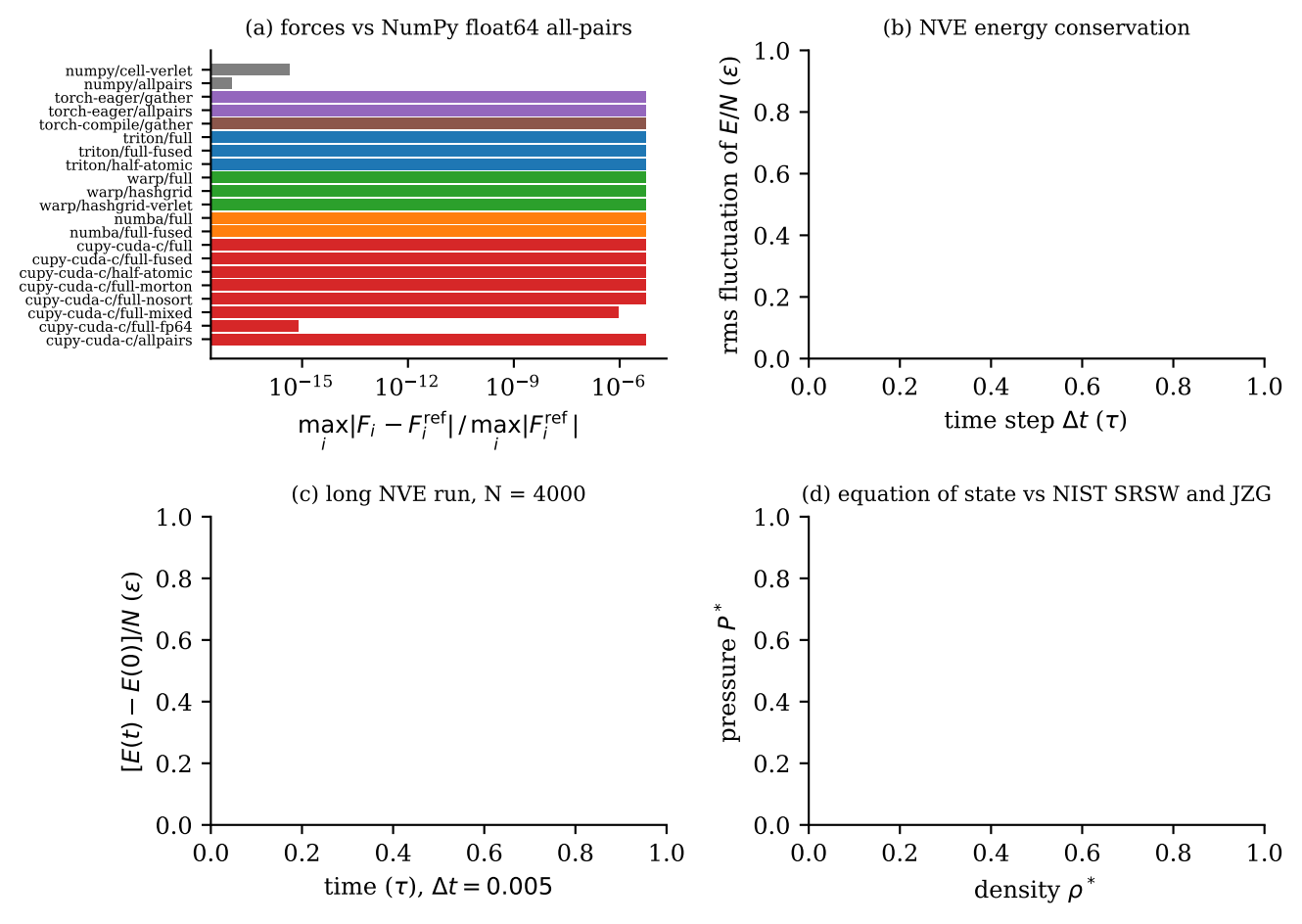
(a) forces vs NumPy float64 all-pairs
numpy/cell-verlet
numpy/allpairs
torch-eager/gather
torch-eager/allpairs
torch-compile/gather
triton/full
triton/full-fused
triton/half-atomic
warp/full
warp/hashgrid
warp/hashgrid-verlet
numba/full
numba/full-fused
cupy-cuda-c/full
cupy-cuda-c/full-fused
cupy-cuda-c/half-atomic
cupy-cuda-c/full-morton
cupy-cuda-c/full-nosort
cupy-cuda-c/full-mixed
cupy-cuda-c/full-fp64
cupy-cuda-c/allpairs
10^{-15}
10^{-12}
10^{-9}
10^{-6}
$\max_i |F_i - F_i^{\mathrm{ref}}| / \max_i |F_i^{\mathrm{ref}}|$
(b) NVE energy conservation
rms fluctuation of E/N (ε)
0.0
0.2
0.4
0.6
0.8
1.0
time step Δt (τ)
(c) long NVE run, N = 4000
$[E(t) - E(0)]/N$ (ε)
time (τ), $\Delta t = 0.005$
(d) equation of state vs NIST SRSW and JZG
pressure P^*
density ρ^*